36 Sub-Sentential Alignment Using Similar Example of Words Correspondences

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1 Introduction

We have proposed a machine-translation method which is based on translation examples and utilizes as many as possible the part-of-speech of words [1]. The corpus required by the system must be aligned at level of words in both sentences. The larger the amount of the translation examples is, the more we are confident in the quality of the experiment results. However, hand-alignment work is time-consuming and therefore not very appropriate for larger texts. In addition, Results from automatic alignment are free of any hesitation over which words correspondences to choose. These hesitations may appear during hand-alignment work. Besides, when the input texts have been modified, by a better version of tokenization program for example, the time-consuming hand-alignment work must be restarted from the begining.

We are tackling a relatively new field in the machine translation. It is the direct French-Japanese spoken-language machine translation. This field is characterized by the non-availability of aligned corpus and appropriate machine-readable bilingual dictionary.

Sub-sentential alignment methods which are based on the co-occurence statistics have already been proposed [3, 4]. They are efficient for large corpus. However, size-limited corpus does not give reliable statistics result. Besides, our research goes beyond a simple extraction of a translation lexicon, and tries to find a set of correspondence maps for each pair of sentences. The results can be used in an example-based machine translation system.

In this paper, we present a semi-automatic method to find words correspondences in corpus. It will be very useful for any experiment in a new field, whose resources are still rarely available. The input corpus are assumed to be tagged and aligned sentence per sentence. Our idea is to use co-occurence statistics to induce an initial translation lexicons. Then, words correspondences which were found in pairs of sentences from these lexicons, are used as a model to find new correspondences in another similar pair of sentences.

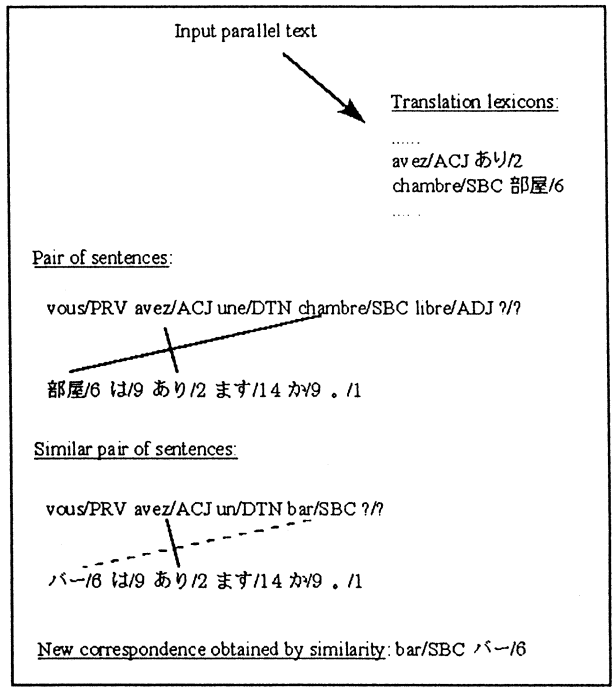


Fig. 1 Overview of the process

2 The Alignment Method

The main goal of the alignment process is to search words correspondences between pair of sentences. Figure 1 shows the overview of the search process. Tokens are in a form *"word/POS[1] tag"*. The search starts by extracting translation lexicons from the corpus. By mapping these translation lexicon to every pair of sentences in the corpus, a model of correspondence maps can be obtained. This model is used to search new word correspondences in another similar sentence.

2.1 Search for an initial translation lexicons

Two tokens co-occur if they occured in an aligned sentence pairs. Since frequent tokens may cause false words correspondences, tokens appearing more than a given number β times in the whole texts are not considered during the count. β was decided by a simple observation of the histogram of words frequencies. The co-occurrence frequency of a word type pair is simply the number of times the pair co-occur in the corpus. As a candidate for the translation lexicons, a pair of tokens

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[1]Part Of Speech

with higher frequency is better than a pair with lower frequency. We assume that no co-occuring two tokens can have a same French or Japanese tokens. Therefore, a pair of tokens will be eliminated from the result if one of its elements has already appeared in a pair of tokens with higher frequencies. The remaining set of pairs of tokens constitutes the translation lexicons.

2.2 Search for a new correspondences

For each pair of French and Japanese sentences (f, j), our method searchs at most 10 other pairs of sentences (fi, ji) where fi or ji match the best respectively f and j. The matching algorithm considers the correspondence between two contiguous POS tags as unit of comparison and searchs a best matching sentence at level of the whole sentence [2]. For each pair of word correspondences (xf, xj) which can be found between (f, j) from the translation lexicons, the algorithms searches if a similar correspondence (xfi, xji) exists in its matching pair sentence (fi, ji). xf, xj, xfi and xji are word positions in the sentence. If $w(i)$ is the word at the position i in a sentence, word correspondences (xf, xj) and (xfi, xji) are similar if the POS tags of $w(xf)$ and $w(xfi)$, as well as $w(xj)$ and $w(xji)$ are equal. If another correspondence (yfi, yji) exists between (fi, ji), and $xfi - \alpha \leq yfi \leq xfi + \alpha$, and $xji - \alpha \leq yji \leq xji + \alpha$, $(yfi - xfi + xf, yji - xji + xj)$ will be a new correspondence if it is similar to (yfi, yji). The same operation is performed for all matching pair of sentences. Many new correspondences are found, some may already existed in the initial lexicons, and some may found many number of times. High frequency correspondences are better that low frequency ones.

The whole process can be run again after adding the correspondences newly obtained and new set of correspondences will be obtained.

3 Experiments and Results

The experiment was done with a corpus taken from French-Japanese conversation books. French texts were tagged by the EBTI tagger program [5] and Japanese texts by Chasen 1.0 [6]. The data is presented in Table 1. During the co-occurrence count, β was 40 for both French and Japanese texts. 40 or higher frequency tokens were therefore not considered. This eliminates 74 tokens (resp. 50 tokens) from the French (resp. Japanese) texts. We observed 100 entries which were taken randomly from the results.

The result of the lexicon extraction is presented in Table 2. 798 entries were extracted at 84.0% of correspondence rate. If only two or higher frequency pair of tokens are considered, 483 entries were extracted at 93.3%.

	French	Japanese
Number of sentences	2753	2753
Total number of tokens	21570	21293
Average sentence length	7.83 words	7.73 words
Max.[2]sentence length	25 words	20 words
Min.[3]sentence length	1 word	1 word
Number of diff.[4]tokens	2957	3321
Number of diff. POS tags	40	14

Table 1 Description of the input texts

	Number of Corresp.[5]	Exact Corresp. Rate
Occurence > 1	482	93.3%
Occurence $= 1$	316	70.0%
Total	798	84.0%

Table 2 Results of the search of translation lexicon

Number of corresp. in the lexicon	896
Number of sentences covered	2415 (87.7%)
Total number of corresp.	5294
Average corresp. per sentence	2.19

Table 3 Input to the search for a new correspondences (after hand-adjustment operation)

	Number of Corresp.	Exact Corresp. Rate	Corresp. newly found
Freq.[6] > 1	583	95.0%	200
Overall	1603	78.0%	812

Table 4 New correspondences

We corrected the translation lexicon by hand and started the search for new correspondences with 896 entries in the lexicons. The input to the second step is presented in Table 3. Table 4 shows the results of the first execution of the process, with $\alpha = 2$ and from the observation of 100 random samples. 812 new correspondences is newly obtained at 78.0%. Among them, 200 can be got at 95.0%. Correct lexicons extraction rate of 93.9% and 95.0% of exact new correspon-

[2]Maximum
[3]Minimum
[4]different
[5]Correspondences
[6]Frequency

dences rate show the effectiveness of our method. The operation can be executed again and again as new correspondences maps are obtained. With a slight handy-correction of the result at every end of cycle, we have now an average of 4.9 correspondences per pair of sentences.

4 Conclusion

We have presented a semi-automatic method for searching word correspondences between bilingual pair of sentences. It is designed for a size-limited texts and in the case where machine readable dictionary are not available. The results showed the effectiveness of the method. Statistics on co-occurrence of tokens within a same sentence would likely improve the method. Checking the result of the whole translation system, however, is a priority for future works.

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